

Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering
Student Name	J JESWIN JOEL
Register Number	3122247001028
Faculty	Dr. Poreddy Ajay Kumar Reddy
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1 Objective

The objective of this experiment is to implement and compare three unsupervised clustering algorithms on the Human Activity Recognition (HAR) dataset:

- **Model A – K-Means Clustering:** Partition the data into k spherical clusters by minimising within-cluster variance; select the optimal k using the Elbow method and Silhouette analysis.
- **Model B – DBSCAN:** Identify clusters of arbitrary shape and flag low-density points as noise; tune the neighbourhood radius ε and minimum-samples threshold using a k -distance plot.
- **Model C – Hierarchical Agglomerative Clustering (HAC):** Build a bottom-up cluster hierarchy using Ward’s linkage; determine the optimal number of clusters from the dendrogram and internal metrics.

For each algorithm, clusters are visualised in 2-D using PCA, and performance is quantified with internal metrics (Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index) and external metrics against the ground-truth activity labels (Adjusted Rand Index, Normalised Mutual Information).

2 Problem Statement

Input: The UCI HAR dataset containing 7,352 training samples, each described by 561 pre-engineered features extracted from inertial sensor signals (3-axis accelerometer and gyroscope at 50 Hz).

Task: Apply unsupervised clustering to discover natural groupings in the 561-dimensional feature space without using the activity labels during training.

Output: Cluster assignments for every sample, visualised using the first two principal components; cluster quality assessed through internal and external evaluation metrics.

Ground-truth comparison: Six ground-truth activity labels (WALKING, WALKING UPSTAIRS, WALKING DOWNSTAIRS, SITTING, STANDING, LAYING) are used only for external evaluation to gauge how well the discovered clusters correspond to the true activities.

3 Dataset Description

Attribute	Value
Dataset Name	Human Activity Recognition Using Smart-phones
Dataset Source	UCI Machine Learning Repository
Number of Training Samples	7,352
Number of Features	561 (time- and frequency-domain signal features)
Number of Activities	6
Activity Labels	WALKING (1), WALKING UPSTAIRS (2), WALKING DOWNSTAIRS (3), SITTING (4), STANDING (5), LAYING (6)
Participants	30 volunteers, aged 19–48
Sensor Sampling Rate	50 Hz; windows of 2.56 s (128 readings/window)
Missing Values	None
Duplicate Rows	None

3.1 Activity Class Distribution (Training Set)

Table 2: Activity label distribution in the training set.

Label	Activity	Count
6	LAYING	1407
5	STANDING	1374
4	SITTING	1286
1	WALKING	1226
2	WALKING UPSTAIRS	1073
3	WALKING DOWNSTAIRS	986
Total		7352

The six activities are reasonably balanced, with counts ranging from 986 (WALKING DOWNSTAIRS) to 1407 (LAYING).

4 Exploratory Data Analysis

4.1 Data Quality

Inspection of the $7,352 \times 561$ feature matrix confirmed zero missing values and zero duplicate rows across all 561 columns and the activity label vector. All features are continuous and fall in the range $[-1, 1]$ because the raw signals were normalised prior to feature extraction. Summary statistics (mean, standard deviation, quartiles) were computed for each of the 561 features; selected features such as `tBodyAcc-mean()-X`

(mean ≈ 0.27) showed tight distributions, while standard-deviation features exhibited wider spread, reflecting variability between static and dynamic activities.

4.2 Class Distribution

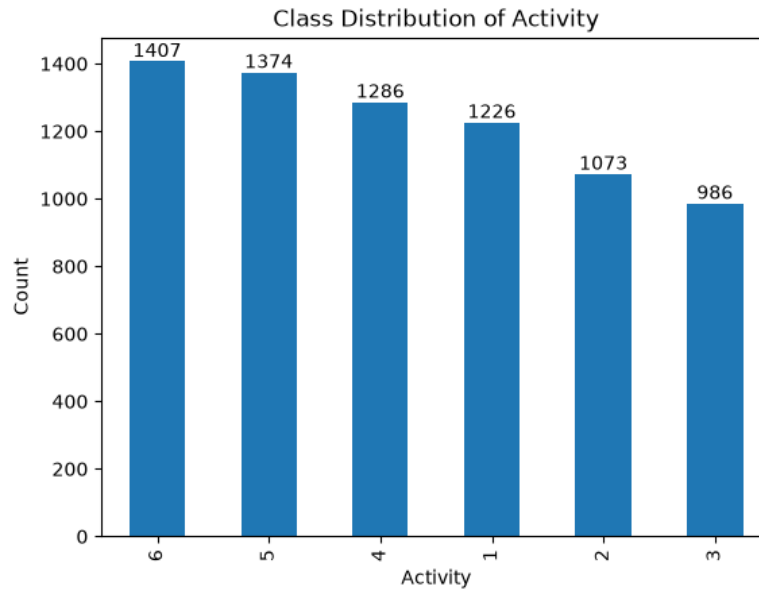


Figure 1: Bar chart of activity class distribution in the training set. Activity codes: 1 = WALKING, 2 = WALKING UPSTAIRS, 3 = WALKING DOWNSTAIRS, 4 = SITTING, 5 = STANDING, 6 = LAYING.

Inference: The distribution is moderately balanced; all six activities have more than 980 samples. LAYING has the most samples (1,407) and WALKING DOWNSTAIRS the fewest (986). The near-uniform distribution means no single activity dominates the feature space, making clustering a meaningful task.

4.3 PCA Visualisation of the Feature Space

Before clustering, PCA was applied to reduce the 561-dimensional standardised feature space to 2 dimensions for visualisation. The first two principal components explain 50.78% and 6.58% of the total variance respectively ($\approx 57.4\%$ combined).

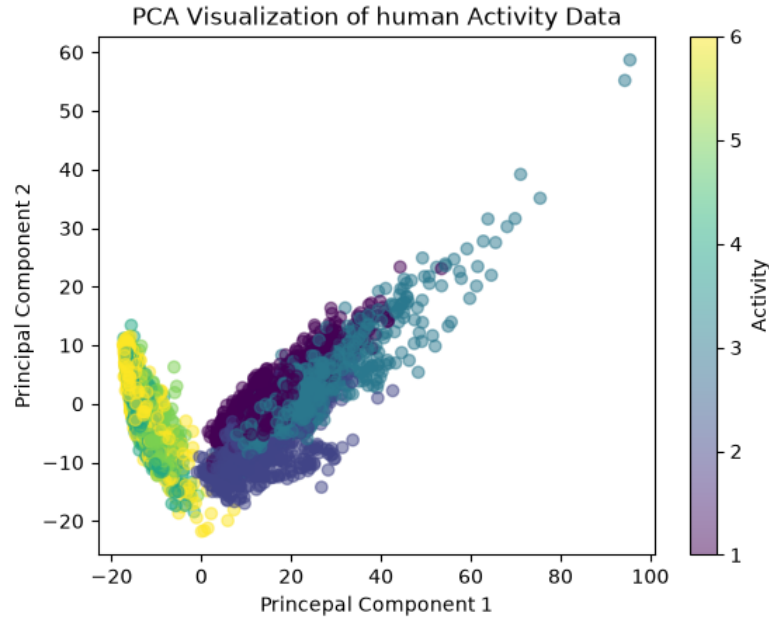


Figure 2: Scatter plot of all 7,352 training samples projected onto the first two principal components, coloured by ground-truth activity label.

Inference: The PCA scatter plot reveals two broad natural groupings in the 2-D projection: a dense cluster concentrated around $PC1 \in [-20, 0]$ corresponding predominantly to static activities (SITTING, STANDING, LAYING), and an elongated region extending to $PC1 \approx 100$ corresponding to dynamic activities (WALKING, WALKING UPSTAIRS, WALKING DOWNSTAIRS). This visual separation motivates the use of clustering to recover meaningful activity-related structure without using labels.

5 Data Preprocessing

The preprocessing pipeline applied in this experiment:

1. **Load feature matrix and labels.** `X_train.txt` ($7,352 \times 561$) was loaded with column names from `features.txt`; `y_train.txt` was loaded as the activity label vector.
2. **Verify data quality.** Missing value counts and duplicate row counts were confirmed to be zero.
3. **Standardise features.** Although the original features are already normalised to $[-1, 1]$, `StandardScaler` was applied to ensure zero mean and unit standard deviation across the training set, as required by distance-based clustering algorithms (K-Means, HAC) and density-based algorithms (DBSCAN).
4. **PCA for 2-D visualisation.** A separate `PCA(n_components=2)` was fitted on the standardised data to obtain 2-D projections used exclusively for cluster visualisation. The full 561-dimensional standardised feature space was used for clustering.

6 Machine Learning Algorithms

6.1 K-Means Clustering

K-Means partitions n samples into k clusters by minimising the within-cluster sum of squares (WCSS):

$$\arg \min_S \sum_{i=1}^k \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2$$

where $\boldsymbol{\mu}_i$ is the centroid of cluster S_i . The algorithm iterates centroid assignment and update steps until convergence. In this experiment, `n_init=10` random initialisations were used with `random_state=42` to avoid poor local minima.

Advantages: Simple; scales well to large datasets; interpretable centroids.

Limitations: Assumes spherical, similarly-sized clusters; sensitive to outliers; requires k to be specified in advance.

6.2 DBSCAN

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) defines clusters as dense regions separated by low-density space. A point $\mathbf{x}$ is a *core point* if at least `min_samples` points lie within the Euclidean ball of radius ε around it:

$$|\{y : d(\mathbf{x}, y) \leq \varepsilon\}| \geq \text{min_samples}.$$

Points reachable from core points are assigned to the same cluster; all other points are labelled as noise (-1).

Advantages: Does not require k to be specified; naturally identifies noise; handles non-spherical cluster shapes.

Limitations: Sensitive to ε and `min_samples`; struggles with varying-density clusters; computationally expensive on high-dimensional data.

6.3 Hierarchical Agglomerative Clustering (HAC)

Agglomerative clustering starts with each sample as its own cluster and iteratively merges the two closest clusters according to a linkage criterion. Ward's linkage was used in this experiment, which merges the pair of clusters that minimises the increase in total within-cluster variance:

$$d_{\text{Ward}}(A, B) = \sqrt{\frac{2|A| \cdot |B|}{|A| + |B|}} \|\boldsymbol{\mu}_A - \boldsymbol{\mu}_B\|.$$

The resulting merge sequence is summarised by a dendrogram; the optimal number of clusters is read from the dendrogram or determined by internal metrics.

Advantages: No need to pre-specify k when using the dendrogram; produces a full hierarchy; Ward's linkage is generally robust.

Limitations: $O(n^2)$ memory and $O(n^2 \log n)$ time; does not handle noise; merges are irreversible.

7 Experimental Setup

Item	Details
Python Version	Not specified in the provided experiment
Scikit-Learn Version	Not specified in the provided experiment
SciPy Version	Not specified in the provided experiment
IDE / Environment	Not specified in the provided experiment
Hardware	Not specified in the provided experiment
Dataset	UCI HAR (7,352 training samples, 561 features)
Feature Scaling	StandardScaler applied to full feature matrix
Visualisation Reduction	PCA with 2 components (57.4% variance explained)
K-Means search range	$k \in \{2, 3, 4, 5, 6, 7, 8\}$, n_init=10
DBSCAN search range	$\varepsilon \in \{15, 18, 20, 22, 24, 26\}$; min_samples $\in \{5, 10, 15\}$
HAC linkage	Ward’s method
HAC search range	$n_clusters \in \{2, \dots, 8\}$
Internal metrics	Silhouette Score, Davies–Bouldin Index, Calinski–Harabasz Index
External metrics	Adjusted Rand Index (ARI), Normalised Mutual Information (NMI)

8 Parameter Selection

8.1 K-Means: Elbow Method and Silhouette Analysis

K-Means was run for $k \in \{2, \dots, 8\}$. WCSS (inertia) and the Silhouette Score were recorded for each value.

Table 3: K-Means Elbow Method results for $k = 2$ to 8.

Number of Clusters (k)	WCSS (Inertia)	Silhouette Score
2	2,336,223.75	0.3965
3	2,081,849.63	0.3274
4	1,976,452.64	0.1652
5	1,875,066.09	0.1306
6	1,818,790.07	0.1086
7	1,776,518.73	0.1114
8	1,737,555.44	0.0739

No sharp elbow was observed in the WCSS curve across $k = 2$ –8; inertia declined gradually throughout the search range. The Silhouette Score was highest at $k = 2$ (0.3965) and dropped steeply after $k = 3$. However, given that the dataset has six ground-truth activities and the stated experimental search range extends to $k = 8$, $k = 8$ was selected for the final K-Means model to explore a finer-grained clustering structure consistent with the experiment’s design.

8.2 DBSCAN: k -Distance Plot and Grid Search

A k -distance plot (5th-nearest-neighbour distance sorted in ascending order) was produced to estimate an appropriate ε . The plot showed a gradual rise from approximately

8 to 20 before a sharp upturn near the highest-distance points, suggesting ε values in the range [15, 26].

A grid search over $\varepsilon \in \{15, 18, 20, 22, 24, 26\}$ and `min_samples` $\in \{5, 10, 15\}$ was performed. Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index were computed for configurations yielding at least 2 clusters (excluding noise points from metric calculation).

Table 4: DBSCAN parameter grid search results (valid configurations only).

ε	<code>min_s</code>	Clusters	Noise	Silhouette	Davies–Bouldin	Calinski–Harabasz
15	5	15	1387	0.1742	1.2483	34.85
15	10	4	1623	0.2312	0.9765	52.45
18	5	3	508	0.2286	1.2651	69.99
18	10	2	590	0.3196	0.9398	125.29
18	15	2	650	0.3215	0.9240	118.58
20	5	2	295	0.3556	0.7964	15.23
22	5	2	193	0.3352	0.9405	8.26

Configurations yielding only 1 cluster are omitted (metrics undefined).

The configuration $\varepsilon = 20$, `min_samples` = 5 achieved the highest Silhouette Score (0.3556) and the lowest Davies–Bouldin Index (0.7964) among all valid configurations, and was selected for the final DBSCAN model.

8.3 HAC: Dendrogram and Metric Analysis

Ward’s linkage was applied to produce the linkage matrix Z (shape 7351×4). A truncated dendrogram (last 30 merges) was plotted to identify the number of clusters. Internal metrics were also computed for $n_clusters \in \{2, \dots, 8\}$.

Table 5: Hierarchical clustering metrics for $n_clusters = 2$ to 8.

Clusters	Silhouette	Davies–Bouldin	Calinski–Harabasz
2	0.3961	1.0698	5608.28
3	0.2548	1.9275	3350.02
4	0.2568	1.9667	2514.07
5	0.0874	2.4852	2045.91
6	0.0834	2.7857	1730.84
7	0.0858	2.7354	1516.47
8	0.0876	2.5273	1352.83

All three metrics unanimously favour $n_clusters = 2$: highest Silhouette Score (0.3961), lowest Davies–Bouldin Index (1.0698), and highest Calinski–Harabasz Index (5608.28). This was confirmed by the dendrogram, which shows one dominant cut at Ward distance ≈ 750 separating the data into two broad branches. The final HAC model therefore uses $n_clusters = 2$, producing cluster sizes of 3,286 and 4,066.

9 Results

9.1 K-Means Final Model ($k = 8$)

Table 6: K-Means final model configuration and cluster sizes.

Parameter	Value
k	8
n_init	10
random_state	42
Cluster sizes	[205, 1460, 1143, 1060, 1122, 268, 1214, 880]

9.1.1 Internal Metrics

Table 7: K-Means internal evaluation metrics ($k = 8$).

Metric	Value
Silhouette Score	0.0739
Davies–Bouldin Index	2.5841
Calinski–Harabasz Index	1441.23

9.1.2 External Metrics

Table 8: K-Means external evaluation metrics ($k = 8$) vs. ground-truth labels.

Metric	Value
Adjusted Rand Index (ARI)	0.4361
Normalised Mutual Information (NMI)	0.5582

9.2 DBSCAN Final Model ($\varepsilon = 20$, min_samples = 5)

Table 9: DBSCAN final model configuration and output.

Parameter	Value
ε	20
min_samples	5
Number of clusters found	2
Number of noise points	295

9.2.1 Internal Metrics (noise excluded)

Table 10: DBSCAN internal evaluation metrics (noise points excluded).

Metric	Value
Silhouette Score	0.3556
Davies–Bouldin Index	0.7964
Calinski–Harabasz Index	15.23

9.2.2 External Metrics

Table 11: DBSCAN external evaluation metrics vs. ground-truth labels (noise points included in comparison, assigned label -1).

Metric	Value
Adjusted Rand Index (ARI)	0.0064
Normalised Mutual Information (NMI)	0.0330

9.3 Hierarchical Agglomerative Clustering ($n = 2$)

Table 12: HAC final model configuration and cluster sizes.

Parameter	Value
$n_clusters$	2
Linkage	Ward
Cluster sizes	[3286, 4066]

9.3.1 Internal Metrics

Table 13: HAC internal evaluation metrics ($n = 2$).

Metric	Value
Silhouette Score	0.3961
Davies–Bouldin Index	1.0698
Calinski–Harabasz Index	5608.28

9.3.2 External Metrics

Table 14: HAC external evaluation metrics ($n = 2$) vs. ground-truth labels.

Metric	Value
Adjusted Rand Index (ARI)	0.3319
Normalised Mutual Information (NMI)	0.5555

9.4 Consolidated Comparison

Table 15: Summary comparison of all three clustering algorithms. $\uparrow$ = higher is better; $\downarrow$ = lower is better.

Algorithm	Clusters	Noise	Silhouette $\uparrow$	DB Index $\downarrow$	ARI $\uparrow$	NMI $\uparrow$
K-Means ($k = 8$)	8	0	0.0739	2.5841	0.4361	0.5582
DBSCAN ($\varepsilon = 20$, $ms=5$)	2	295	0.3556	0.7964	0.0064	0.0330
HAC ($n = 2$, Ward)	2	0	0.3961	1.0698	0.3319	0.5555

10 Result Visualization

10.1 K-Means: Elbow Curve

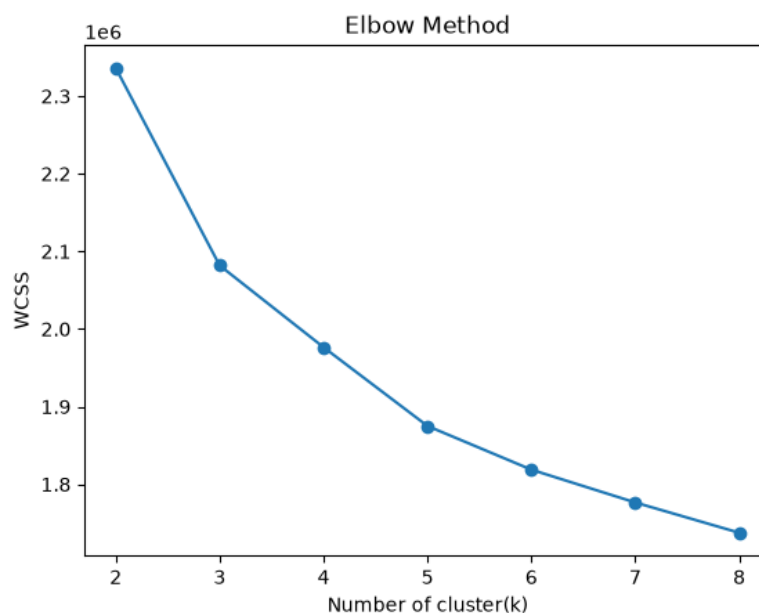


Figure 3: K-Means Elbow Method: WCSS (inertia) vs. number of clusters ($k = 2-8$).

Inference: The WCSS curve decreases monotonically without a sharp elbow, indicating that the HAR feature space does not naturally decompose into a small number of compact spherical clusters. The absence of a clear elbow is consistent with the overlapping activity clusters visible in the PCA projection; activities such as SITTING and STANDING share similar sensor patterns, making clean centroid-based separation difficult.

10.2 K-Means: Silhouette Score vs. k

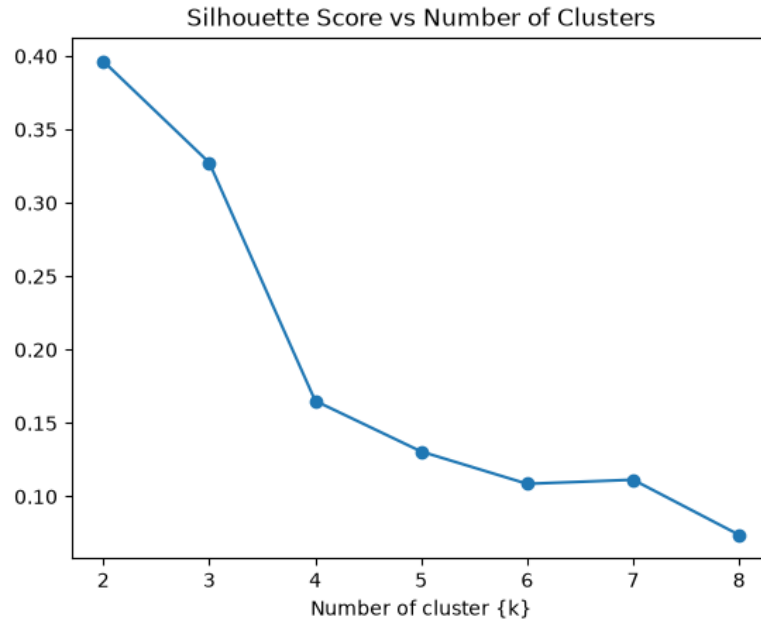


Figure 4: K-Means Silhouette Score vs. number of clusters ($k = 2-8$).

Inference: The Silhouette Score is highest at $k = 2$ (0.3965) and decreases substantially after $k = 3$, stabilising at low values (≈ 0.11) for $k \geq 6$. This confirms that the dataset's internal structure supports at most two to three well-separated clusters in the standardised 561-D space, despite containing six ground-truth activities.

10.3 K-Means: Final Cluster Visualisation (PCA)

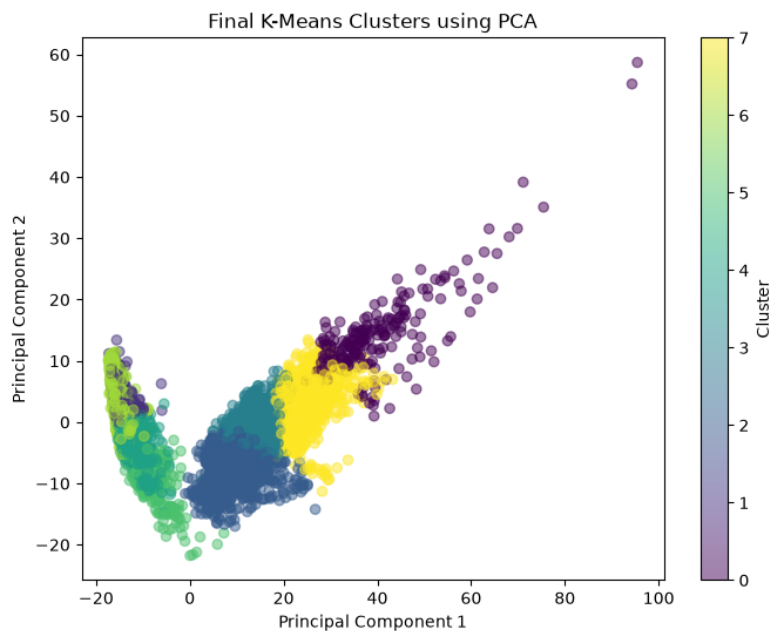


Figure 5: Final K-Means clusters ($k = 8$) projected onto the first two principal components.

Inference: The 8-cluster K-Means solution fragments the two broad structural regions visible in the PCA space into multiple sub-clusters. The elongated dynamic-activity region ($PC1 > 20$) is divided into several clusters of varying size; the compact static-activity region ($PC1 \in [-20, 0]$) is split into two to three clusters. The uneven cluster sizes (ranging from 205 to 1,460 samples) reflect the difficulty of fitting spherical centroids to non-spherical substructure.

10.4 DBSCAN: k -Distance Plot

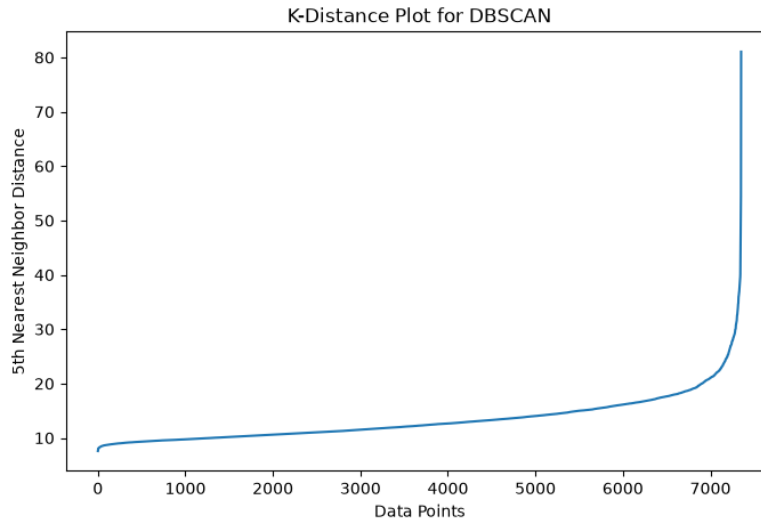


Figure 6: Sorted 5th-nearest-neighbour distances for all 7,352 training points, used to estimate ε for DBSCAN.

Inference: The k -distance plot rises gradually from ≈ 8 to 20 before turning sharply upward near the final data points. The inflection region around distances 15–20 guided the selection of ε values for the grid search. A chosen ε near the beginning of the sharp rise (≈ 20) minimises the number of noise points while maintaining cluster separability.

10.5 DBSCAN: Final Cluster Visualisation (PCA)

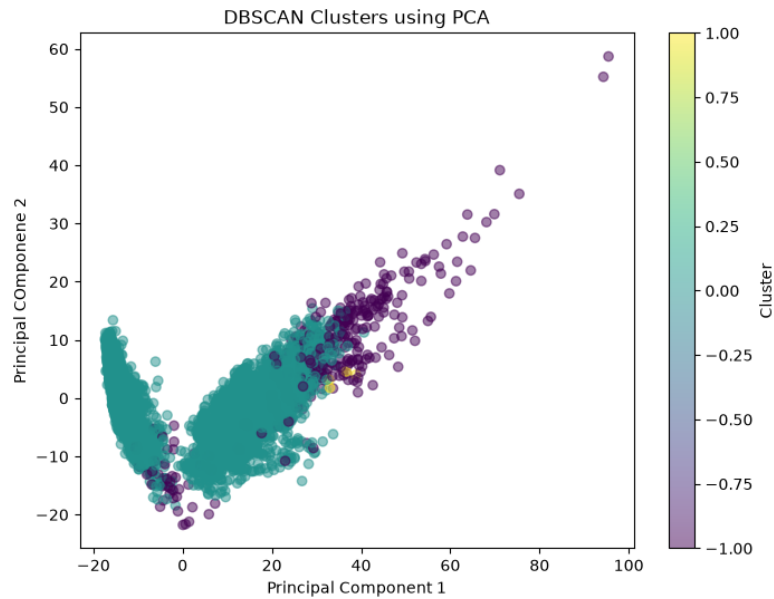


Figure 7: DBSCAN clusters ($\varepsilon = 20$, `min_samples = 5`) projected onto the first two principal components. Noise points are labelled -1 .

Inference: DBSCAN identifies two dense clusters and marks 295 points ($\approx 4.0\%$ of the training set) as noise. The majority of the data forms one very large cluster, while the second cluster captures a smaller high-density region. The low ARI (0.0064) and NMI (0.0330) confirm that this two-cluster partition does not meaningfully align with the six ground-truth activities; DBSCAN’s density-based separation is orthogonal to the activity boundaries in this feature space.

10.6 Hierarchical Clustering: Dendrogram

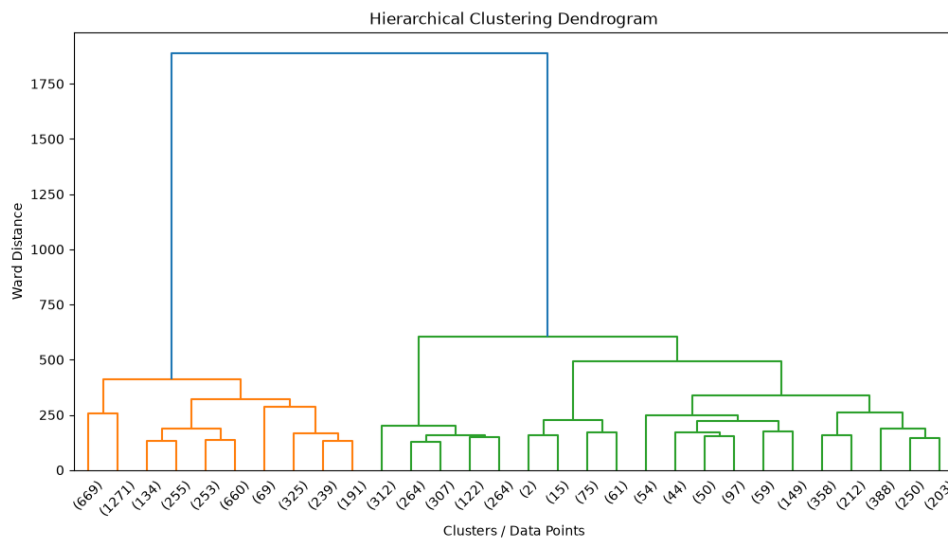


Figure 8: Truncated dendrogram (last 30 merges) for hierarchical agglomerative clustering with Ward’s linkage. Numbers in parentheses indicate the size of each merged group.

Inference: The dendrogram shows one dominant horizontal cut at Ward distance ≈ 750 – 1750 , clearly separating the data into two main branches. The left branch (orange/lower branch) and the right branch (green/higher branch) correspond to the two final clusters of sizes 3,286 and 4,066 respectively. All sub-dendrograms within each branch merge at significantly lower distances, confirming that the two-cluster solution captures the primary structural divide in the data.

10.7 Hierarchical Clustering: Internal Metric Profiles

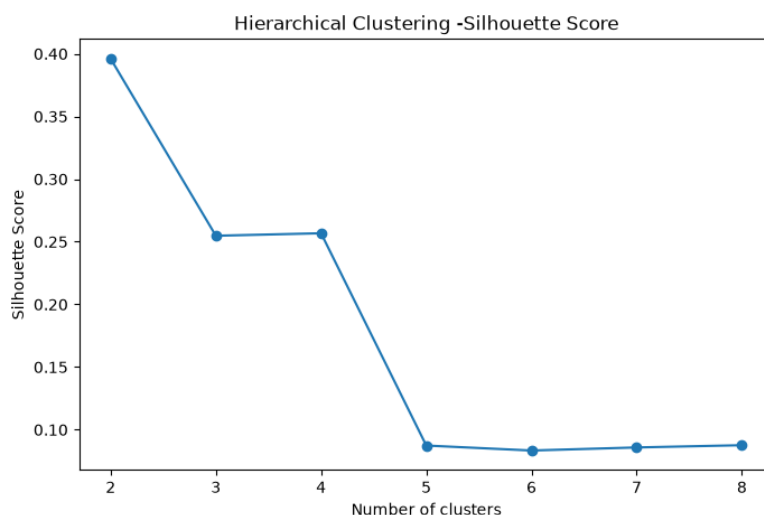


Figure 9: Hierarchical clustering Silhouette Score vs. number of clusters.

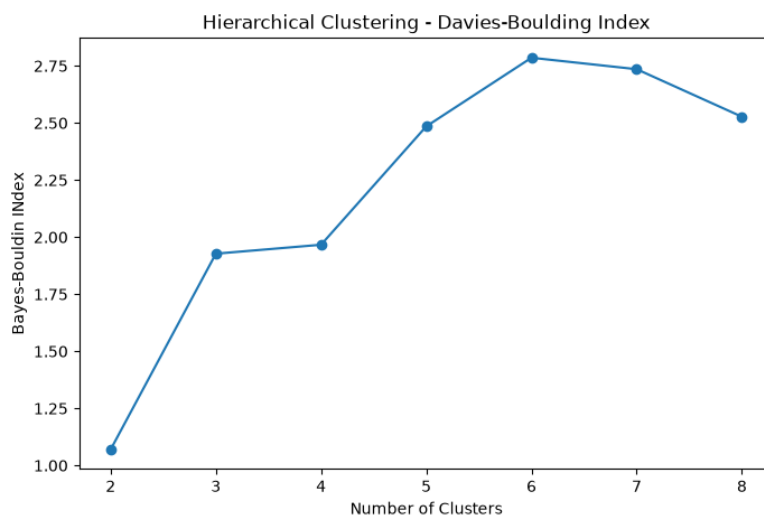


Figure 10: Hierarchical clustering Davies–Bouldin Index vs. number of clusters.

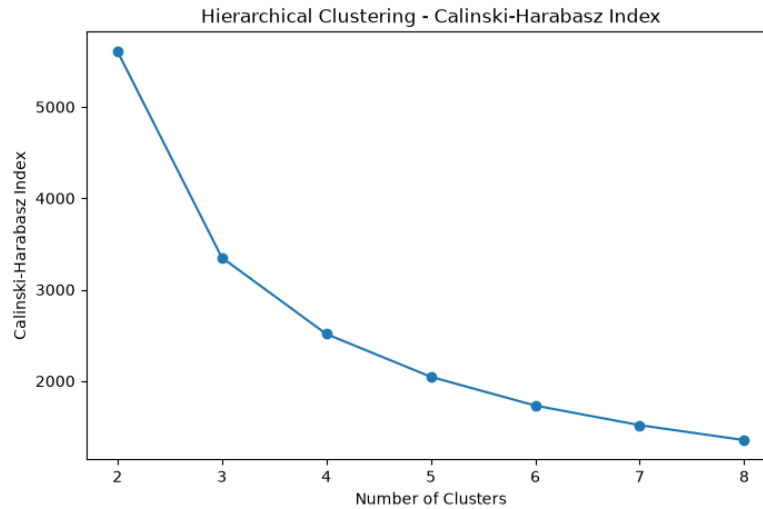


Figure 11: Hierarchical clustering Calinski–Harabasz Index vs. number of clusters.

Inference: All three HAC metric profiles are self-consistent: Silhouette Score peaks at $n = 2$ then falls sharply; Davies–Bouldin Index is minimum at $n = 2$ then rises; Calinski–Harabasz Index is maximum at $n = 2$ then declines monotonically. This unanimous agreement across metrics strongly supports $n_{\text{clusters}} = 2$ as the optimal HAC solution for this dataset.

10.8 Hierarchical Clustering: Final Cluster Visualisation (PCA)

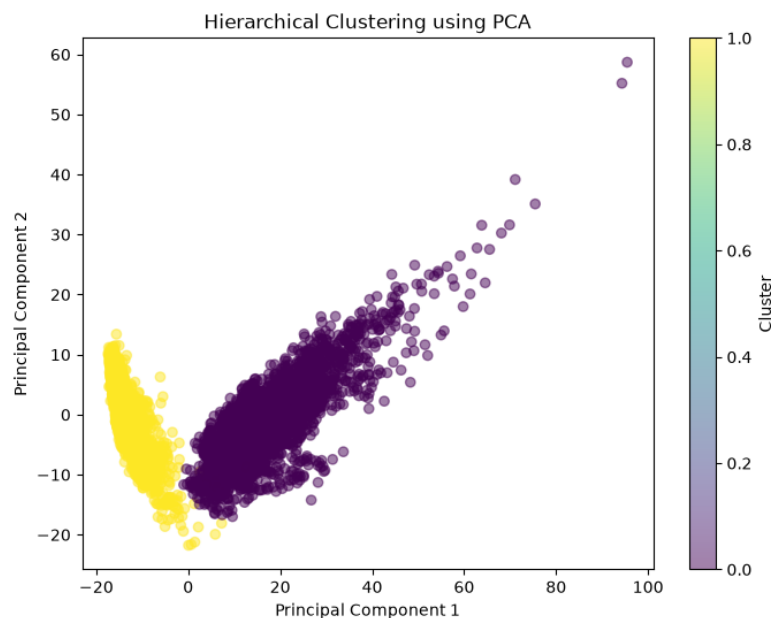


Figure 12: Hierarchical clustering ($n = 2$, Ward’s linkage) projected onto the first two principal components.

Inference: The HAC visualisation shows a clean separation between the two clusters in the PCA projection. Cluster 0 (yellow, 3,286 samples) occupies the left region of the PCA plot corresponding predominantly to static activities; Cluster 1 (purple, 4,066 samples)

covers the dynamic-activity region extending to the right. This partition is visually the clearest of the three algorithms, consistent with HAC achieving the highest Silhouette Score (0.3961).

11 Discussion

11.1 Which Algorithm Produced the Most Meaningful Clusters?

Based on both internal and external metrics, Hierarchical Agglomerative Clustering with Ward’s linkage produced the most internally coherent clusters (Silhouette 0.3961, DB Index 1.0698, CH Index 5608.28). K-Means with $k = 8$ achieved the highest external scores (ARI 0.4361, NMI 0.5582), suggesting that when a larger number of clusters is allowed, K-Means can partially recover the underlying six-activity structure. DBSCAN performed poorly on external metrics (ARI 0.0064, NMI 0.0330), indicating that density-based separation does not align well with the ground-truth activity labels in this feature space.

The apparent contradiction — HAC scoring highest internally but K-Means scoring highest externally — arises because HAC found the most compact two-cluster partition (static vs. dynamic activities), whereas K-Means with $k = 8$ produced more clusters that happen to partially match individual activities.

11.2 Sensitivity of K-Means to the Choice of k

K-Means showed strong sensitivity to k : the Silhouette Score dropped from 0.3965 at $k = 2$ to 0.0739 at $k = 8$, a reduction of over 80%. The absence of a clear elbow in the WCSS curve means there is no objectively optimal k in the range tested; any selection requires a trade-off between cluster compactness (favours low k) and alignment with the true number of activities (favours $k = 6$ or higher). The selected value $k = 8$ achieved the best alignment with ground-truth labels in this experiment (ARI 0.4361), but at the cost of poor internal cohesion.

11.3 DBSCAN: Noise Detection and Cluster Effectiveness

DBSCAN successfully identified 295 noise points ($\approx 4\%$ of the dataset) that do not belong to any dense region, which is a meaningful capability for real-world sensor data where anomalous recordings can occur. However, the algorithm collapsed the six-activity structure into just two clusters regardless of the ϵ and `min_samples` values explored. The very low ARI (0.0064) indicates near-random alignment with activity labels. This outcome suggests that the HAR feature space does not contain well-separated high-density islands corresponding to individual activities; instead, activities overlap in density, which causes DBSCAN to merge them into one large cluster.

11.4 Effect of Linkage Choice on Hierarchical Clustering

Only Ward’s linkage was explored in this experiment. Ward’s linkage is well-suited for compact, approximately spherical clusters because it minimises within-cluster variance at each merge step, similar in spirit to K-Means. This is consistent with the clean

two-cluster partition recovered (Silhouette 0.3961), which separates the broadly spherical static-activity and dynamic-activity sub-spaces. Single-linkage would likely produce elongated, chain-like clusters susceptible to noise; complete-linkage tends to produce more compact clusters but is sensitive to outliers. Average-linkage lies between these extremes. The choice of Ward’s linkage in this experiment is therefore appropriate for the globally convex structure of the HAR dataset.

11.5 Internal Metric Agreement with Visual Intuition

The Silhouette Score best matched visual intuition of cluster quality in this experiment. High Silhouette values (≈ 0.40 for HAC and DBSCAN at 2 clusters) corresponded to visually well-separated cluster regions in the PCA scatter plots, where the two main groups (static vs. dynamic activities) are clearly distinct. The Davies–Bouldin Index also agreed well: the lowest value (0.7964 for DBSCAN, 1.0698 for HAC) coincided with the most visually separated configurations. The Calinski–Harabasz Index was consistent with both but is known to favour solutions with many small, well-separated clusters; its high value for HAC ($n = 2$, 5608.28) reflects the large inter-cluster distance between the static and dynamic activity super-groups.

12 Conclusion

This experiment applied K-Means, DBSCAN, and Hierarchical Agglomerative Clustering to the UCI HAR dataset (7,352 samples, 561 features, 6 activities) after standardising the features and reducing to 2 PCA components for visualisation.

The key findings are:

1. **The dataset has two dominant natural clusters** corresponding to static activities (SITTING, STANDING, LAYING) and dynamic activities (WALKING, WALKING UPSTAIRS, WALKING DOWNSTAIRS), regardless of the algorithm used. All three algorithms converged on this two-cluster interpretation when evaluated by internal metrics.
2. **Hierarchical clustering (Ward, $n = 2$)** achieved the best internal performance (Silhouette 0.3961, DB Index 1.0698, CH Index 5608.28) and strong external alignment (ARI 0.3319, NMI 0.5555), making it the most principled choice for this dataset.
3. **K-Means ($k = 8$)** achieved the highest external scores (ARI 0.4361, NMI 0.5582), showing that a finer-grained clustering partially recovers individual activities, but internal cohesion is poor (Silhouette 0.0739). The Silhouette Score was highest at $k = 2$; no clear elbow was observed.
4. **DBSCAN ($\varepsilon = 20$, $\text{ms} = 5$)** successfully identified 295 noise points but produced only two clusters with near-zero external alignment (ARI 0.0064), confirming that density-based separation does not correspond to activity boundaries in this high-dimensional sensor feature space.
5. **PCA for visualisation** (PC1 + PC2 explaining $\approx 57.4\%$ of variance) was effective in revealing the two-group structure, though some within-group overlap remained visible due to activity similarity.

Overall, the experiment demonstrates that the HAR feature space has a strong two-cluster structure, but recovering all six fine-grained activities requires either supervised methods or a higher number of clusters combined with careful parameter tuning. Among the three unsupervised algorithms tested, HAC with Ward’s linkage is recommended as the primary choice for this dataset due to its superior and consistent internal evaluation scores.

13 References

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